Caterham Seven Suspension

A Buyer's Guide to Specifying Coilovers Properly

By Simon Rogers · Too Fast To Race · Updated 2026

Why I Wrote This

For years — back when I ran RaceShocks.UK and Meteor Motorsport — I'd sit down every few months and rewrite a buyer's guide like this one. The cars changed, the products changed, and so did my own thinking. This is that guide, brought up to date for Too Fast To Race and for where we are now.

The biggest change is what we've chosen to leave out. Over the years we sold a wide range of brands and specifications, including our own CORE (by TracTive) label and the Quantum ranges. After a great deal of testing, building and living with these dampers on real Caterham's, we've made a deliberate decision: we now concentrate on the two manufacturers that, in our hands, consistently deliver the best balance of performance, serviceability and value for the Seven — Nitron and Penske. Narrowing the range isn't about offering less choice; it's about offering the right choice, backed by deep knowledge of two platforms rather than shallow knowledge of many.

Everything else that mattered in the old guide still holds. The core principle hasn't changed: we'll always find the best product for your use and your budget. That is the heart of how we work — and it's why we'd like to see you come back to us as the car develops, and tell your friends while you're at it.

What You're Actually Buying

The real value in a kit from us isn't access to a damper. Anyone can sell you a damper. The value is in how it's specified for your car.

What determines how your Caterham behaves — on the road or the circuit — is not the number of adjusters on the damper. It's whether the spring rates and the valving have been chosen for your actual car, your tyres and the way you drive. A correctly specified single-adjustable kit will comprehensively out-drive a more adjustable kit that's been set up generically. The hardware sets the ceiling; the specification decides whether you ever reach it.

That's the whole point of talking to me rather than ordering from a manufacturer who knows the damper but not the car. Every Penske kit we supply, for instance, is built on valving I've developed and proven over years of real Caterham competition — front and rear digressive/linear pistons on our 3 way 8700 kits. Chosen deliberately for how the Seven loads each axle. You won't find that in a catalogue.

And here's something many people don't expect: you often feel the quality of a good damper more on the road than on the track. The circuit is smooth and consistent; the road is not. But on track there's a stopwatch — and good suspension setup gives you confidence, and confidence is what produces lap time. The gains show up as a car that's easier to drive, more neutral, less edgy, with longer and more even tyre wear.

How to Choose

Don't choose by price alone, and don't assume more adjustment is automatically better. Choose by where your car sits and how you use it:

- Fast road, track days, road-and-track or Nürburgring trips → **Nitron NTR R1.** For the overwhelming majority of Caterhams, this is the right answer.
- A car you're actively developing — adjusting between events with intent → **Nitron NTR R3.**
- A committed programme that values Penske build quality and rebuildability → **Penske 7500 COSA or CODA.**
- A serious, data-led race programme → **Penske 8700 3-Way.**

If you're not sure, that's exactly the conversation to have with us — and we'll give you a straight answer for your car.

The Range at a Glance

Kit	Adjustment	From (+VAT)
Nitron NTR R1 40mm	Single	**£1,579**
Nitron NTR R3 40mm	Compression & rebound	**£3,206**
Penske 7500 COSA	Single	**£3,392**
Penske 7500 CODA	Compression & rebound	**£3,758**
Penske 8700 3-Way	HS comp / LS comp / rebound	**£7,156**

All prices are "from", excluding VAT and shipping. Every kit is specified for the Caterham Seven across the S3, SV and CSR chassis and comes with mounting spacers; the specification work behind it is what adapts the kit to your particular car.

The Five Kits

Nitron NTR R1 40mm from £1,579 +VAT

Single-adjustable. One adjuster controls compression and rebound together, fine-tuning the car around a base that has already been calibrated for the application. On a Seven that single adjuster isn't a compromise — it's exactly what most road and track drivers actually need.

This is the kit for fast road cars, track day cars, road-and-track compromise builds: a composed, communicative, genuinely fast car, set correctly once and used with confidence. Specified properly, it will out-drive plenty of more expensive, more adjustable kits that were never set up correctly.

My take — ***for most people reading this, the R1 is the answer — and the best use of your budget.***

Nitron NTR R3 40mm from £3,206 +VAT

Independently adjustable compression and rebound. These are genuinely different events with different requirements, and the R3 lets you address one without disturbing the other.

That capability is only an advantage in the right hands. Used to refine a setup against chassis behaviour you can identify and measure, it's a precision instrument. Used without a process, the extra adjustment simply gives you more ways to move away from a setting that was already right. The hardware doesn't make the difference — the method does. However do remember we are always here to help and as part of our sales process we will help you understand how and what to adjust. Its actually rather more simple than you think.

This is the kit for the dedicated owner: circuit and hillclimb cars under active development, drivers logging data and adjusting with a clear objective, builds refined progressively across seasons.

My take — ***if your car is a fast road or occasional track day car, the R1 is almost certainly the better choice, and a better use of budget. I'll tell you honestly which side of that line your car sits on.***

Penske 7500 COSA   from £3,392 +VAT (clicker from £3,636)

The single-adjustable entry to genuine Penske ownership. The same manufacturing precision, consistency and full rebuildability Penske is trusted for across world motorsport, valved specifically for the Seven. A single external adjuster fine-tunes the car around a base that's already correct.

Every COSA kit we build runs my own special valving — front digressive/linear, rear linear. It's offered with two adjuster styles: the standard sweep adjuster, set with a pin tool (the robust, proven choice, from £3,392), or the clicker eyelet for quicker, repeatable adjustment by hand, which many competition drivers prefer (from £3,636). The kit is otherwise identical — the choice is purely the adjuster.

My take — ***this is for the committed driver who wants genuine Penske build quality and the ability to rebuild and re-valve as the car develops. For many fast road and lighter track day cars, a correctly specified R1 delivers outstanding performance for considerably less — and I'll say so honestly.***

Penske 7500 CODA   from £3,758 +VAT (clicker from £4,002)

The double-adjustable 7500. Same platform as the COSA, with a second external adjuster so compression and rebound can be set independently — without the complexity or packaging of a remote reservoir. Similar valving, same deliberate front digressive/linear, rear linear configuration, now with the freedom to refine each direction as the car evolves.

My take — ***the kit for the Penske owner who has a process and adjusts with purpose, and who values the rebuildability for long-term development. If you want most of that benefit for less, the Nitron R3 gives you independent adjustment at a lower price — I'll give you an honest comparison.***

Penske 8700 3-Way   from £7,156 +VAT

The top of the range, and the most capable damper we offer for the Seven. A triple-adjustable, remote-canister coilover that separates high-speed compression, low-speed compression and rebound into three independent adjustments. The remote canister is what makes true high-speed compression control possible — managing how the damper responds to sharp surface inputs such as kerbs and sharp bumps, independently of how it controls body movement.

On the Caterham 8700 that's 17 clicks of high-speed compression, 22 of low-speed compression and 80 of rebound. Both front and rear run digressive/linear pistons, chosen for a programme that will use the full breadth of that adjustment purposefully across both axles. I run this damper myself.

That's a great deal of adjustment, and it's an incredible advantage in the hands of someone with the process to use it. This is the kit for a genuine track or race use — conditions that vary between events, and a budget that reflects a serious long-term commitment. For most Caterhams it's more damper than the application justifies, and I'll tell you that plainly. But once you have used a set its very very difficult to go back.

My take — ***the way I'd describe the difference at this level: with a Penske you can feel the grains of sand on the road; with most other dampers you feel the small stones.***

Springs — What Comes With Your Kit, and What to Upgrade To

Every Nitron kit comes with Nitron springs as standard; every Penske kit comes with Eibach. Both are good springs. The question is whether to upgrade.

Option	Applies to	Cost (+VAT)
Eibach springs (vs Nitron standard)	Nitron kits	**+£200**
Hyperco springs (vs Nitron standard)	Nitron kits	**+£280**
Hyperco springs (vs Eibach standard)	Penske kits	**+£120**
Hyperco hydraulic spring platforms	All kits	**£175 each***
Torrington bearings (set of 4)	All kits, unless on hydraulic platforms	**£39 ea · £156/kit**

*** The hydraulic spring platforms include shipping *only* when you're also upgrading to Hyperco springs; otherwise add shipping. All option prices exclude VAT.**

Why consider Hyperco? Hyperco springs are manufactured to a higher specification than any other coilover spring — rated more accurately, tested at more positions, and carrying a lifetime warranty no one else offers. They're the exclusive spring supplier to the NASCAR Cup Series, which means every Next Gen Cup car runs Hyperco springs as a matter of the rulebook. Hyperco's own record is that their springs have been under every winner of the Indianapolis 500 for more than fifty consecutive years — and when IMSA's current hybrid prototypes were introduced, every car in the field ran Hyperco.If you want the most consistent, most precisely rated spring under your car, this is it.

Linear, across the board. I run linear springs on every kit, front and rear — and that's a deliberate decision. A linear spring holds the same rate wherever the car is in its travel, so the feedback through the chassis stays consistent and predictable: you learn the car once, and it behaves the same way every time you drive it. Progressive springs have their place, but their rate changes as they compress, and that variability works against the clarity I want you to feel. I'd rather control ride compliance with the damper — a digressive piston on the front takes the harshness out of sharp bumps and kerbs while keeping strong low-speed control — and leave the spring doing one honest, predictable job. The result is a car that talks to you the same way in every corner.

Pistons & Valving — Why "Digressive" Matters

This is the part that genuinely separates a good kit from an ordinary one, and it's worth understanding.

Think of a damper as a timing device. It controls the speed at which the chassis transfers its weight. A damper with different forces at different velocities lets you influence that timing — and that's the whole game.

In a linear piston, the damping force rises in a straight line: gentle at low damper speeds, much firmer at high speeds (potholes, kerbs). A digressive piston blends the force off at the high-speed end, so the car isn't harsh when it hits sharp bumps and kerbs — while keeping strong control at the lower speeds, which is where you actually feel the car. I prefer a double-digressive arrangement: the rebound is digressive too, so the spring can extend quickly under high loads to keep the tyre tracking the surface. The result is strong feedback and confidence when you're driving, without the brittleness that puts people off stiff setups. On the Seven, our preferred configuration is front digressive/linear and rear linear — chosen for how the car loads each axle.

Ride Height — Hydraulic Platforms or Bearings

Adjusting ride height on a coilover means turning the spring platform, and friction in that joint makes it harder than it should be. Two ways to solve it:

- **Hyperco hydraulic spring platforms** — the premium solution, £175 each. Add shipping only if you're not also upgrading to Hyperco springs; if you are, it's already included.
- **Torrington bearings** — the value solution, £39 each, four per kit (£156 +VAT per kit). They reduce friction so the platforms turn freely. I'd recommend these on every kit that isn't running the hydraulic platforms.

What Would I Do?

With no budget set it's hard to choose for you — so tell me your figures and I'll guide you accurately. But broadly:

- **The sweet spot for most people:** Nitron NTR R1, specified properly — £1,579 +VAT. It out-drives most things and leaves budget for tyres and track time.
- **Stepping up to independent adjustment:** Nitron NTR R3 (£3,206) if you'll use it with a process, or the Penske 7500 COSA (£3,392) for genuine Penske ownership and rebuildability.
- **Ultimate:** Penske 8700 3-Way (£7,156) for a serious race programme — or the 7500 CODA (£3,758) for most of the benefit at far less cost.

The only thing I'll add is what I always have: I offer a stiffer setup for the track day owner and a more forgiving one for road and road-and-track. That's the whole idea of speaking to me rather than to a manufacturer who knows the damper but not the car.

Upgrading From Standard Dampers

If you're still on the dampers the car came with, here's the honest picture. Standard and OE-style performance dampers are built for broad road use and long life, not for a light, fast Seven being driven hard. They're typically a little under-damped for the way you now drive — not quite enough force to control the car's momentum — and they can vary from unit to unit. They're rarely "failed"; they're simply doing a different job. Any correctly specified performance kit will transform the car's ride, control and precision — and as long as we get the spring rates right, it won't be harsh. If you want to know exactly where yours stand, they can be dyno tested.

Lead Times

Current lead time is approximately 6 weeks across all kits. Every kit is built and tested for the individual car, which is part of why it arrives right.

Let’s Specify Yours

Tell me about your Caterham — chassis (S3, SV or CSR), your tyres, and how and where you use the car. I’ll recommend the right kit honestly, specify it to suit your car rather than just sell it to you, and give you a straight answer if something simpler suits you better.

Start your specification: toofasttorace.com/contact